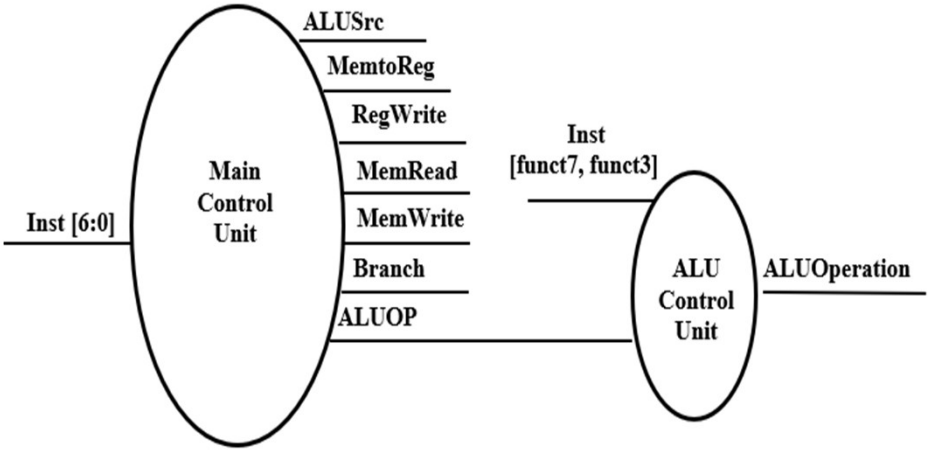
1. Main Control Unit



1

Designing Main Control Unit

Instruction	opcode						Control Signals								
	op_6	op_5	op_4	op_3	op_2	op_1	op_0	1: ALUSrc	2: MemtoReg	3: RegWrite	4: MemRead	5: MemWrite	6: Branch	7: ALUop1	8: ALUop0
R-Type	0	1	1	0	0	0	1	0	0	1	0	0	0	1	0
Lw	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0
Sw	0	1	0	0	0	1	1	1	X	0	0	1	0	0	0
Beq	1	1	0	0	0	1	1	0	X	0	0	0	1	0	1

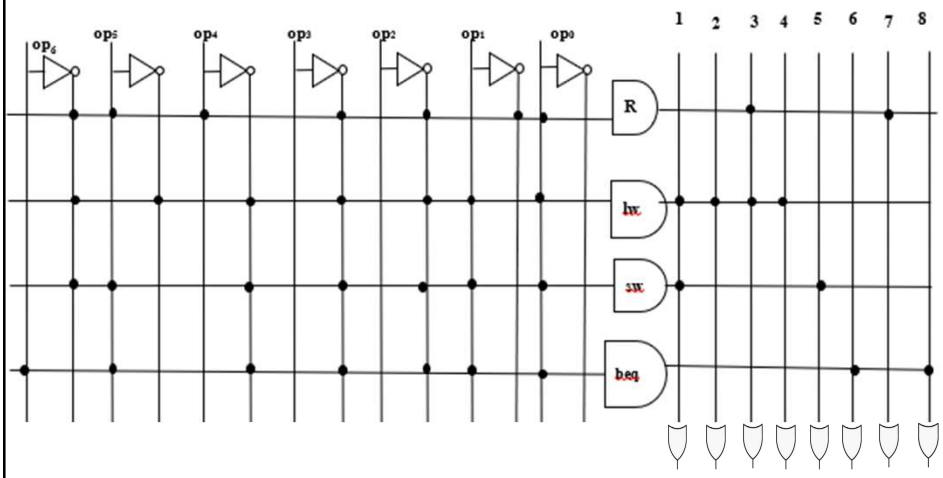
2

Recall …

Signal	Asserted [1]	Deasserted [0]
ALUSrc	When second operand comes from sign-extension unit (lw/sw)	When second operand comes from second register (R-type / branch)
MemtoReg	When value to be stored to register is coming from Memory (lw)	When value to be stored in register is coming from ALU result (R- type)
PCSrc	When PC = BTA	PC = PC+4

3

PLA Implementation of Main Control Unit



3: **ALUSrc = op0**

4

Class Activity

- Write Simplified equations for above 8 control signals.

RegWrite =

MemWrite =

B =

ALUSrc =

MemtoReg =

ALUop0 =

MemRead =

ALUop1 =

7

Why SCDP is not used today?

- Because it is inefficient in terms of time, power consumption, utilization of modules and hardware cost
- Every instruction has same cycle length
- Clock period is determined by the slowest instruction (lw in our design – it uses 5 units) penalizing faster instruction
- Each functional unit can be used only once per clock; therefore, some functional units must be duplicated, raising the cost of the implementation
- All modules whether needed or not are working increasing the power consumption.

8

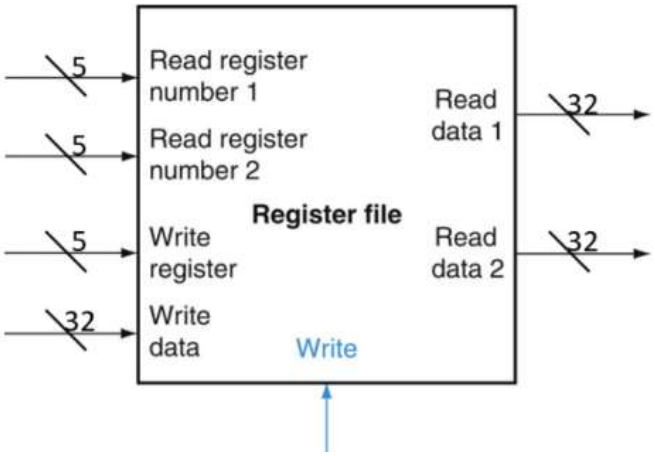
Why SCDP is not used today?

- Circuit timings has to be considered properly.
- For complex systems e.g. processor involving floating point arithmetic- this doesn't work well at all.
- It violates the design principle "make common case fast" , as making the common case fast is useless, if this common case is not the worst case. Here worst case needs to be improved
- Modern processors use pipelining to overcome this.
- Overall performance is poor – but good for simple processors with small instruction sets.

9

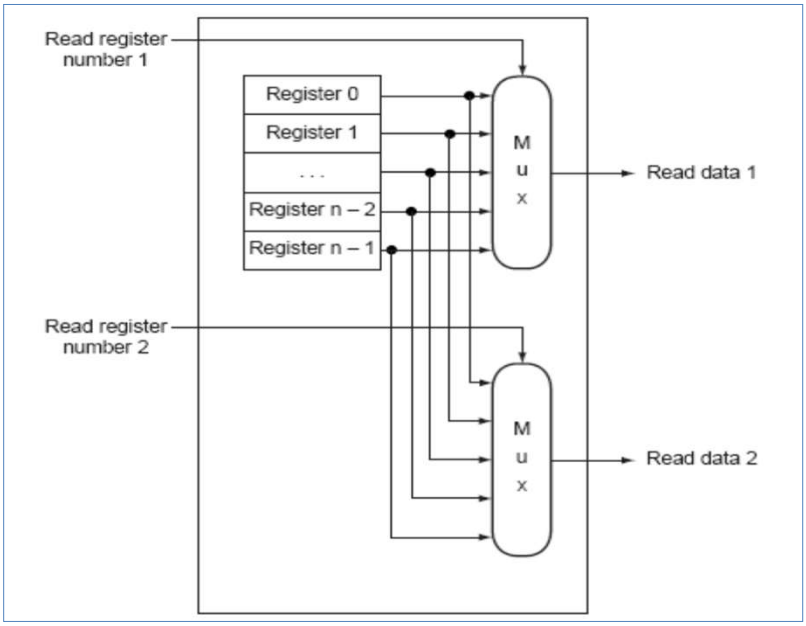
Register File

Note: *No* **Read** *Signal is needed*

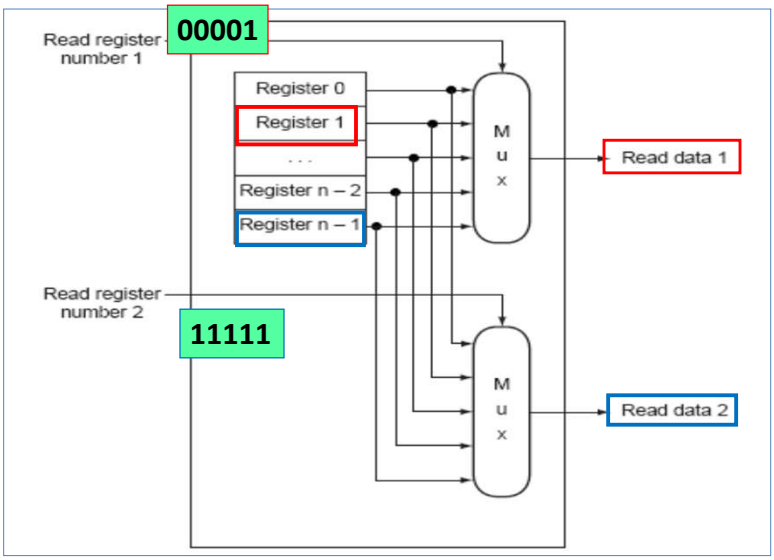


10

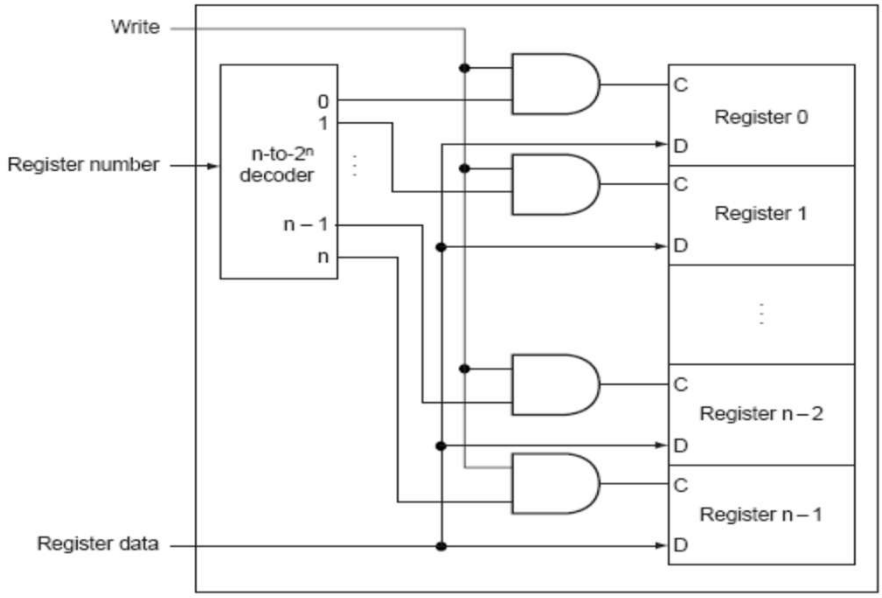
Implementing Read Part



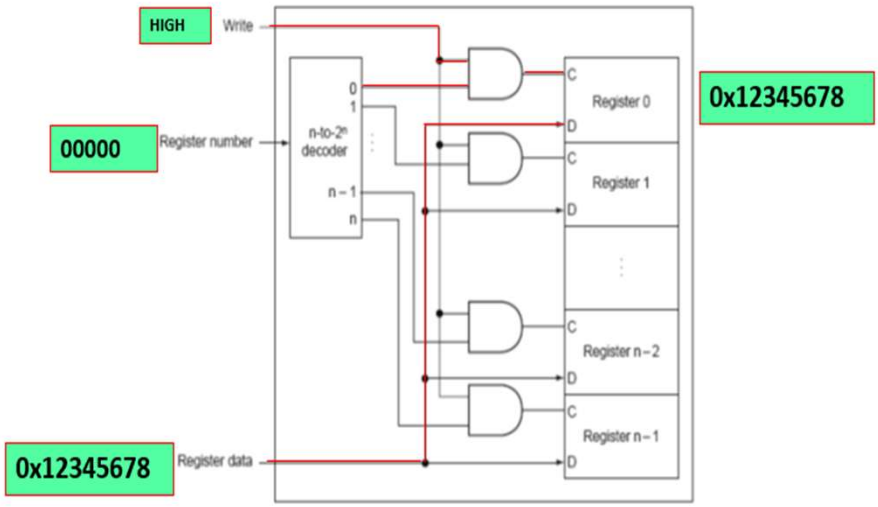
Implementing Read Part



Implementing Write Part



Implementing Write Part



Modified SCDP

